○ E-Commerce Data Analysis Report

Name :- Sanket Nitin Borade

1. Executive Summary

In this dataset analysis, we studied the operational and financial performance of an ecommerce business using customer, product, location, and transaction data.

We cleaned the data, combined different datasets, explored the data, and calculated important KPIs.

This helped us understand revenue trends, find operational risks, and identify opportunities for business growth.

2. Objective of the Analysis

Analyse customer, product, and transaction datasets.

Compute key business KPIs (Revenue, Profit, Return Rate, CLV).

Identify performance patterns and operational risks.

Provide data-driven strategic recommendations

2. Data Overview & Methodology

Data Sources: "C:\Users\Hp\Downloads\project\Assessment – Dataset & Submission Guidelines.zip"

- Customers Data
- Products Data
- Transactions Data
- Location Data

Methodology:

- Data cleaning (missing values, duplicates, formatting corrections)

- Dataset merging using validated relational keys
- Feature engineering (revenue, profit, margin)
- KPI computation
- Exploratory data analysis (trend & imbalance detection)

3. Data Description

- Customers dataset: customer demographic information
- Products dataset: product and pricing details
- Transactions dataset: sales and order records
- Locations dataset: geographic information

4. Data Preparation

- Find the Missing value
- Find the Duplicate values and remove it
- Check the Formatting correction
- Merging the Dataset using primary–foreign key relationships

The given dataset was not in clean format firstly find the null values from the all dataset . there are some null values so replace it . then check the unique values from the columns. see the duplicated values from the columns . then check the formatting of all the columns and find the outliers from all the dataset .

5. Financial Performance Overview

- Total Revenue: ₹ 30127071628
- Region wise revenue :- North 3078061404

  South 2589019162

  West 24459991062

- Total Profit: ₹ 74388902

The total revenue from all the dataset is ₹ 30127071628 and I find

The region wise revenue .

The total profit from all the dataset is ₹ 74388902.

6. Customer Analytics

Total Unique Customers: 5439

Customer Lifetime Value (CLV): ₹ 5581154.432752872

The total unique customers is 5439

And the customer lifetime values is ₹ 5581154.432752872

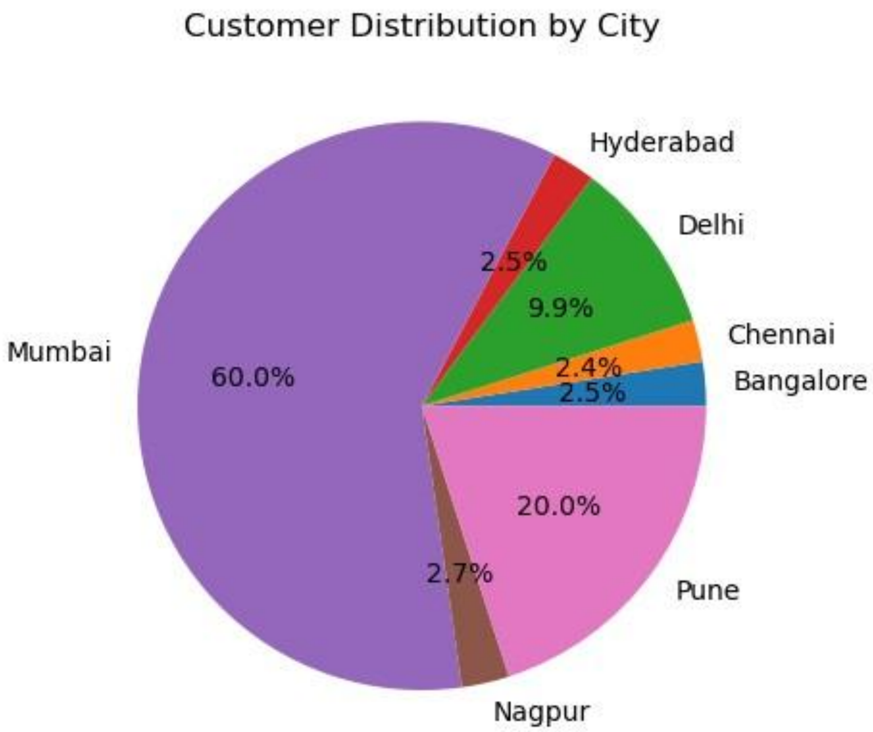
7. Operational Performance Return Analysis

- Total Return order : 3819
- Total orders : 25460
- Total return rate : 15%

The total return orders is 3819 and the total orders is 25460 .

The return rate is 15%.

8. City-wise Customer Distribution %



9. Payment Method Distribution

UPI 12730

Card 7638

COD 3819

Wallet 1273

This payment method are used to by the product .

10. Key Risks Identified

- There are risk in Revenue concentration
- There are High return rate risk
- There are Regional imbalance risk
- There are Margin compression in certain product segments.

Key Insights

- Most of the sales are coming from a few products and cities.
- Some products are being returned more often, which is reducing profit.
- A small number of customers are spending more money than others.
- Some cities have lower sales compared to others.

11. Recommendations

- Increase marketing efforts in cities or regions where sales are low to improve performance.
- Improve product quality and check delivery processes carefully for items that have a high return rate.
- Create special offers and loyalty programs to retain customers who spend more and contribute higher revenue.
- Focus more on selling products that generate higher profit instead of only high sales volume.

12. Final Conclusion

This analysis shows that the business is earning good revenue.
But there are some risks, like depending too much on a few customers and some operational issues.

To grow steadily in the future, the company should:

- Find more customers (not depend on only a few),
- Keep existing customers happy,
- Improve daily operations and processes.